

Q1.

- (a) Use Simpson's rule, with five ordinates (four strips), to calculate an estimate for

$$\int_0^4 \frac{x}{x^2 + 2} dx$$

Give your answer to four significant figures.

(4)

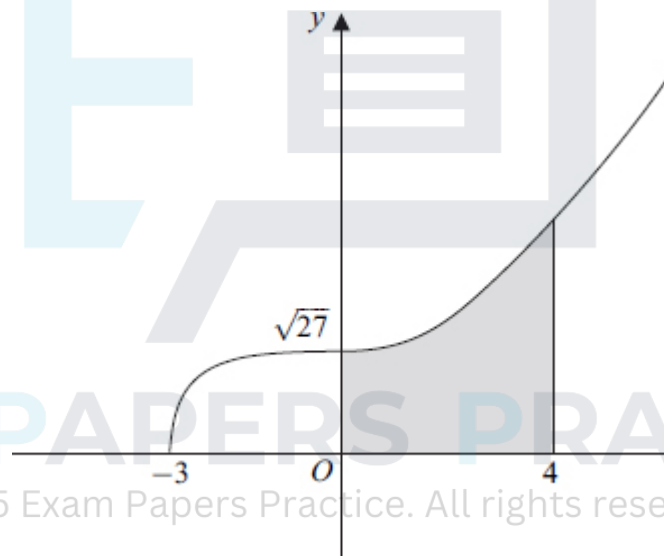
- (b) Show that the exact value of $\int_0^4 \frac{x}{x^2 + 2} dx$ is $\ln k$, where k is an integer.

(5)

(Total 9 marks)

Q2.

The diagram shows a sketch of the graph of $y = \sqrt{27 + x^3}$.



- (a) The area of the shaded region, bounded by the curve, the x -axis and the lines $x = 0$

and $x = 4$, is given by $\int_0^4 \sqrt{27 + x^3} dx$.

Use the mid-ordinate rule with **five** strips to find an estimate for this area. Give your answer to three significant figures.

(4)

- (b) With the aid of a diagram, explain whether the mid-ordinate rule applied in part (a) gives an estimate which is smaller than or greater than the area of the shaded region.

(2)

(Total 6 marks)

Q3.

Use Simpson's rule with 7 ordinates (6 strips) to find an estimate for

$$\int_0^3 4^x dx.$$

(Total 4 marks)

Q4.

Use the mid-ordinate rule with four strips to find an estimate for $\int_{0.4}^{1.2} \cot(x^2) dx$, giving your answer to three decimal places.

(Total 4 marks)

Q5.

Use the mid-ordinate rule with four strips to find an estimate for $\int_0^{0.4} \cos\sqrt{3x+1} dx$, giving your answer to three significant figures.

(Total 4 marks)

Q6.

(a) Use the mid-ordinate rule with four strips to find an estimate for $\int_0^{12} \ln(x^2 + 5) dx$, giving your answer to three significant figures.

(4)

(b) A curve has equation $y = \ln(x^2 + 5)$.

(i) Show that this equation can be rewritten as $x^2 = e^y - 5$.

(1)

(ii) The region bounded by the curve, the lines $y = 5$ and $y = 10$ and the y -axis is rotated through 360° about the y -axis. Find the exact value of the volume of the solid generated.

(4)

(Total 9 marks)

Q7.

(a) Use Simpson's rule with 7 ordinates (6 strips) to find an approximation to

$$\int_{0.5}^2 \frac{x}{1+x^3} dx,$$

giving your answer to three significant figures.

(4)

(b) Find the exact value of $\int_0^1 \frac{x^2}{1+x^3} dx$.

(4)

(Total 8 marks)

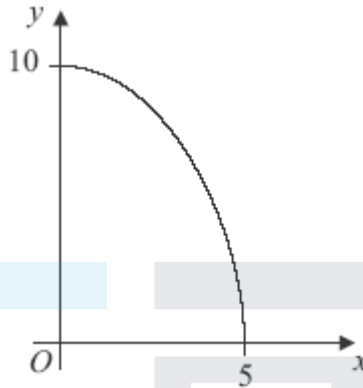
Q8.

Use Simpson's rule with 5 ordinates (4 strips) to find an approximation to $\int_1^9 \frac{1}{1+\sqrt{x}} dx$, giving your answer to three significant figures.

(Total 4 marks)

Q9.

The diagram shows the curve with equation $y = \sqrt{100 - 4x^2}$, where $x \geq 0$.



- (a) Calculate the volume of the solid generated when the region bounded by the curve shown above and the coordinate axes is rotated through 360° about the **y-axis**, giving your answer in terms of π .

(5)

- (b) Use the mid-ordinate rule with five strips of equal width to find an estimate for $\int_0^5 \sqrt{100 - 4x^2} dx$, giving your answer to three significant figures.

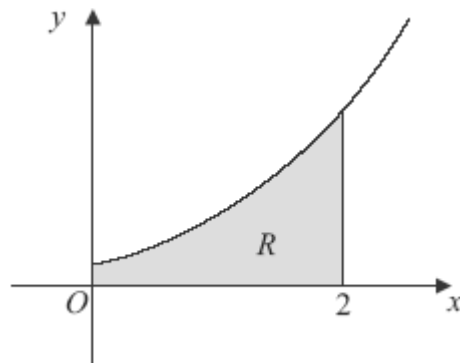
(4)

(Total 9 marks)

©2025 Exam Papers Practice. All rights reserved.

Q10.

The diagram shows the curve with equation $y = (e^{3x} + 1)^{\frac{1}{2}}$ for $x \geq 0$.



- (a) Find the gradient of the curve $y = (e^{3x} + 1)^{\frac{1}{2}}$ at the point where $x = \ln 2$.

(5)

- (b) Use the mid-ordinate rule with four strips to find an estimate for $\int_0^2 (e^{3x} + 1) \frac{1}{2} dx$, giving your answer to three significant figures.

(4)

- (c) The shaded region R is bounded by the curve, the lines $x = 0$, $x = 2$ and the x -axis. Find the exact value of the volume of the solid generated when the region R is rotated through 360° about the x -axis.

(4)

(Total 13 marks)

Q11.

Use the mid-ordinate rule with four strips of equal width to find an estimate for

$$\int_1^5 \frac{1}{1 + \ln x} dx$$

, giving your answer to three significant figures.

(Total 4 marks)

Q12.

Use Simpson's rule with 5 ordinates (4 strips) to find an approximation to

$$\int_1^3 \frac{1}{\sqrt{1+x^3}} dx$$

giving your answer to three significant figures.

(Total 4 marks)

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.